

A BPMN-based Automated Approach for the Analysis of Healthcare Processes

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Abstract—Healthcare organizations are increasingly pushed to improve the quality of care service taking into account the increasing complexity in patient treatment and the continuous reduction of available resources. The adoption of Business Process Management (BPM) practices is thus becoming a key enabler for the improvement of healthcare processes (HPs). Accordingly, methods and tools are required to address behavioral and performance aspects from the early phases of the process lifecycle in order to improve the quality of healthcare, reduce costly reworks and increase the effectiveness of BPM approaches. This paper specifically addresses the specification and analysis phases of the process lifecycle and introduces a model-driven method for healthcare process simulation. The proposed method is based on a model transformation approach that takes as input the process specification in BPMN, appropriately extended to include the performance properties of the process, and yields as output the corresponding process simulation code, ready to be executed. In order to illustrate the method and its effectiveness, the paper describes an example application to a process dealing with the hip fracture for elderly patients.

Keywords—BPMN, Healthcare Process, Business Process Management, process simulation, model transformation.

I. INTRODUCTION AND RELATED WORK

Healthcare organizations are increasingly pushed to improve the quality of care services in an unfavourable context, which is characterized by the increasing complexity in patient treatment and the continuous reduction of available resources [1].

Huge efforts have been devoted to develop and implement evidence-based practices with the objective of standardizing and improving healthcare quality and efficiency, yet a considerable variation in performance still persists [2].

Recent studies show how most of the healthcare failures are of operational type, due to processes that are not appropriately designed for a reliable and efficient delivery of care [3]. As a consequence, researchers and practitioners increasingly advocate process redesign as a valuable way to reduce practice variation and improve the quality of care, while at the same time taking into account efficiency issues [4].

In this respect, the adoption of *Business Process Management (BPM)* practices is increasingly becoming a key enabler

for the improvement of organizational performance. According to well known BPM frameworks [5], the *healthcare process (HP) lifecycle* includes the following main phases: (i) definition (ii) specification, (iii) analysis; (iv) implementation, (v) execution and monitoring and (vi) diagnosis and improvement.

The specification and analysis phases are critical to carry out an effective process redesign, because they allow business analysts to define and predict the process performance before committing the resources required for the actual implementation [6]. Accordingly, methods and tools are required to analyze the *behavioural* and *performance* aspects from the early stages of the *HP lifecycle*, in order to improve the quality of healthcare, reduce costly reworks and increase the effectiveness of BPM approaches.

In this respect, the use of modeling and simulation approaches has proven to be essential to quantitatively assess the operational performance and assist the decision making process, either before or after the process implementation [7].

The use of these techniques in healthcare management has founded to be promising for the improvement of the outcome and efficiency of care, because they allow to identify and test different potential service designs through quantitative evidence-informed analysis, which takes into account quality and safety issues, as well as the cost impact [8].

However, their extensive use in healthcare is still limited if compared to other industries [9]. As outlined in [10] the effective adoption of simulation-based analysis techniques is still hindered by some substantial limitations, mainly deriving by the fact that simulation models are often too complex to be used by people having no technical background and their parameterization requires a vast amount of data which often are difficult to collect and analyze for simulation purposes.

These limitations are particularly stressed in the healthcare domain, because of various factors: (i) healthcare processes are highly complex, distributed and multi-disciplinary [1], [11]; (ii) medical knowledge is often tacit and rapidly evolving [12]; (iii) domain knowledge is essential to successfully implement process-oriented management practices [13]; (iv) the adoption of information technologies is still in its early stages if compared to other industries and data, when available, are

often disseminated in many different information systems (IS) [14]; (v) the staff of healthcare organizations often deficiencies of the modelling and analysis skills [9].

As a consequence of the aforementioned issues, literature and empirical findings point out that the communication gap among domain experts (clinical staff) and technical people (process analysts) [15] and lack of complete and reliable data are the main barriers to the application of modeling and simulation techniques in healthcare.

Therefore, the development of methods and tools to support the full involvement of the process stakeholders in the definition of the formal process models needed to carry out quantitative analysis is advocated. In order to address the requirement for a deeper stakeholders engagement in the process specification and analysis, an increasing attention is directed toward the development of fully-facilitated modeling and simulation approaches [16].

In the present work we propose a model-driven method for healthcare process modeling and simulation, which is based on the use of BPMN (Business Process Model & Notation) to specify the healthcare process. The proposed method exploits a lightweight BPMN extension that is applied to annotate the BPMN model of the healthcare process and automate the production of the corresponding simulation model, which is finally executed to get the results that support the process analysis activities.

The reminder of this paper is structured as follows: Section II briefly introduces the use of BPMN for the process specification, as well as the BPMN extension and the simulation platform at the basis of the proposed method, which is overviewed in Section III. The method application is illustrated in Section IV, by use an example process dealing with the hip fracture for patients over 65. Finally, Section V gives some concluding remarks.

II. BACKGROUND

A. Healthcare process specification

This section reviews the proposed approach for the specification of the *behavioural* and *performance* aspects of processes. As mentioned in the previous section, the main challenges to be faced for the successful implementation of BPM approaches in healthcare concerns the requirement of flexibility and the need for a full involvement of stakeholders in the different phases of *HP lifecycle*.

Therefore, we propose a lightweight approach to *HP specification*, which aims to be from one side flexible enough to be reactive to changes and to be used also by non-technical people, and from the other side semi-structured, to support both quantitative process analysis and process implementation.

The *behavioral aspects* are specified by defining the *HP workflow*, i.e., the orchestration of a set of tasks executed by human or automated resources in order to deliver healthcare services.

The proposed approach exploits a well-known BPM enabler, specifically BPMN (Business Process Model & Notation), which is considered the de-facto standard for specifying business process models [17]. BPMN models can be easily

understood by all (technical and non-technical) stakeholders of HPs.

Several contributions apply BPMN in the context of healthcare [7]. Clinical pathways are an effective approach to manage medical practice in a process-oriented way. Unstructured approaches (e.g. text-based approaches, service blueprinting) for documenting clinical pathways does not allow automatic analysis and makes the models ineffective [18]. BPMN can satisfy the requirements of clinical pathway models for creating semantically annotated diagrams of such pathways and is able to manage more effectively variability, which is one of the main problems to face for effectively dealing with HPs.

As a matter of fact, the primary objective of BPMN is to support the communication between business users, which own the domain knowledge but have no technical expertise, and the technical people, mainly devoted to process analysis and software development. This important feature together with the capability to model complex and distributed processes as well as communication patterns have made BPMN a promising notation for the specification of complex processes, such as *HPs*, and to support their further analysis and enactment through IT platforms [20].

The *performance* aspects refer to the definition of process constraints and Key Performance Indicators (KPI) metrics. The early definition of performance indicators in the specification phase eases the parameterization of simulation models and the identification of data collection procedures for process monitoring purposes.

Unfortunately, BPMN does not support the process characterization process in terms of performance properties. Standard BPMN models do not contain neither the specification of overall performance and reliability constraints, nor the specification of task properties.

To overcome such limitations, the proposed approach makes use of PyBPMN (Performability-enabled BPMN), a lightweight BPMN extension that addresses the specification of performance properties of business processes, as described in the next section.

B. Performance-based annotation of BPMN models

The primary objective of BPMN is to provide a standard notation easily understood by the different stakeholders involved in the specification, implementation, and maintenance of business processes, from the business analysts who specify processes to the technical developers who implement the process execution platform, down to the business people who are responsible for managing and monitoring those processes.

As aforementioned, BPMN only deals with the functional characterization of processes, without providing the ability of specifying non-functional properties, such as performance and reliability [21]. To overcome such limitations, we have introduced PyBPMN, a lightweight BPMN extension for the performability-oriented annotation of BP diagrams [22] [23].

PyBPMN exploits the concept of performability, defined as a joint measure of performance and reliability. In this paper case, the use of PyBPMN is limited to the performance characterization of processes, which are annotated with performance

properties as follows (according to a standard profiling-based extension mechanism):

```
<<property name>> attribute name =
{tag1=value1, tag2=value, ...}
```

As an example, to specify that a given task requires 200 minutes to be executed, the following annotation is specified for to the corresponding task instance:

```
<<PaService>> serviceTime = {value=200,
unit=m, source=est, dir=decr, statQ=max}
```

where the tags *value*, *unit*, *source*, *dir* and *statQ* give the value, the unit of measure, the origin, the type of order relation used to compare values and the type of statistical value of the attribute, respectively. Interested readers are sent to [22] and [23] for a complete description of PyBPMN.

BPMN models annotated by use of PyBPMN include the data that are required to build and parameterize relevant simulation models. In order to ease the simulation model building and parameterization activities, the approach proposed in this paper exploits the *eBPMN* language, which is described in the next section.

C. *eBPMN*: a language for process simulation

eBPMN is a domain-specific language that has been designed and implemented according to the BPMN 2.0 execution semantics, thus obtaining a compact and precise executable model of the input BP model, without the need of introducing additional formalisms, such as queueing-based modeling languages [24].

eBPMN is built on top of *SimArch*, a layered software architecture that provides a general-purpose and event-based simulation infrastructure upon which domain-specific simulation components can be implemented and seamlessly used in local or distributed execution environments [26].

In a BPMN model, a sequence flow is defined as a set of flow elements that are executed according to a well defined process workflow. The occurrence of a process start event acts as a trigger for instantiating and starting the process. The *token* concept is introduced in *eBPMN* to simulate the execution of a business process. More specifically, the occurrence of a start event generates a token on its outgoing sequence flow. The token then passes through the sequence flow edges and routes throughout the different *eBPMN* elements corresponding to the BPMN elements, according to the process structure and to the execution semantics.

eBPMN provides a set of primitives that can be easily mapped to the standard primitives of BPMN. The proposed model-driven method allows to automatically derive the executable *eBPMN* code by use of automated model transformations, without being required to know and manually use the language syntax. Therefore, it can be transparently adopted also by HP stakeholders who are not familiar with simulation tools and languages.

The *eBPMN*-based HP simulation requires a parameterization activity that is carried out by use of the PyBPMN-based annotation of BPMN models. Interested readers are referred to [24] for a detailed description of the *eBPMN* syntax and semantics.

III. OVERVIEW OF THE MODEL-DRIVEN METHOD FOR HEALTHCARE PROCESS SIMULATION

The proposed method exploits the principles introduced by model-driven development and MDA (Model Driven Architecture), which support the development of software systems through the transformation of abstract models to executable components and applications. MDA emphasizes the role of models as the primary artifacts of development by providing a set of guidelines for structuring specifications expressed as models and the *transformations* between such models. The transformation maps the elements of a source model to the elements of a target model that is specified in terms of the same or of a different language [19].

The proposed model-driven method for the simulation-based *HP* analysis provides an effective support along the whole *HP* lifecycle, from the *HP* specification down to the *HP* execution, monitoring and improvement. This section describes its application to the *specification* and *analysis* phases by use of a reference scenario dealing with a specific *HP*, namely the *hip fracture* process.

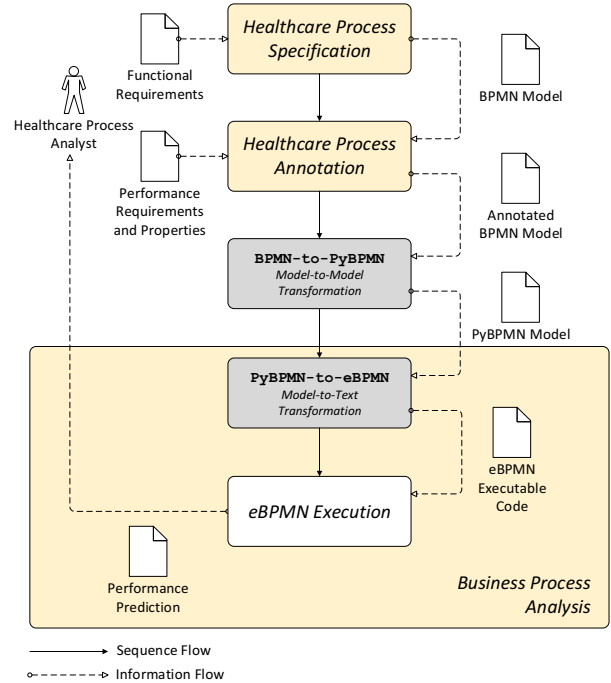


Fig. 1. Model-driven method for healthcare process simulation.

Figure 1 illustrates the proposed method, which consists of the following steps:

- **Healthcare Process Specification:** at this step the *HP* is specified by the relevant process analyst in terms of a standard BPMN model, which takes into account the specific functional requirements in terms of tasks and their interconnections.
- **Healthcare Process Annotation:** the BPMN model obtained from the previous step is enhanced with

textual annotations describing performance requirements of the *HP*, in terms of relevant KPIs, and the task performance properties. The resulting model is referred to as the *Annotated BPMN model* in Figure 1.

- *BPMN-to-PyBPMN Transformation*: this step takes as input the *Annotated BPMN model* produced in the previous step and maps the BPMN annotations to the corresponding PyBPMN elements, thus yielding as output the PyBPMN model. Such transformation is specified in QVT/Operational Mappings, the standard language for the specification of *model-to-model* transformations, provided by the Object Management Group (OMG) as part of the MDA effort [19].
- *PyBPMN-to-eBPMN Transformation*: this *model-to-text* transformation takes as input the *PyBPMN model* and yields as output the *eBPMN Executable Code*. The *PyBPMN-to-eBPMN* transformation is specified in Aceleo, the OMG standard language for the specification of *model-to-text* transformations.
- *eBPMN Execution*: in this final step, the *eBPMN executable code* (i.e., the simulation code) is executed and the *Performance Prediction* is obtained. The process analyst is thus enabled to use the feedback provided by such prediction and compare various alternatives by simply varying the non-functional parameters specified in the *Business Process Annotation* step.

The last two steps of the method are part of the *HP analysis* activity illustrated in Figure 1. Such activity allows process analysts to check whether or not the relevant KPIs are met by the specified *HP* and, in the negative case, plan corrective actions.

Additional details about the *BPMN-to-PyBPMN* and *PyBPMN-to-eBPMN* transformations can be found in [23].

IV. REFERENCE EXAMPLE

This section illustrates the method described in Section III by use of an example application to the acute care of elderly (i.e., over 65) patients with hip fracture diagnosis.

This is an example of process for which international and national clinical-evidence guidelines providing evidence of the link between recommendations and outcome of care have been developed and are readily accessible. In such a case, clinical process indicators can be reliably used to evaluate and improve the *HP* performances [27].

Figure 2 illustrates the BPMN model of the hip fracture process. The process starts with the arrival of elderly patients with suspected hip fracture to the *Emergency Department (ED)*.

If the physical examination and the X-Ray results confirm the hip fracture diagnosis, the patient is admitted to the *Orthopaedic Ward (OW)* where pre-operation activities are carried out. At the time planned for the operation the patient is transferred to the *Operating Theatre* in order to undergo the surgical intervention.

Next, the patient can return directly to the *Orthopaedic Ward*. Alternatively, in case of need, the patient is treated in

the *Intensive Care Unit (ICU)* and then go back to the *OW*, where post-operation activities are performed before discharge.

A. Specification

The *HP definition* consists of the informal description of the general clinical guidelines for elderly patients with hip fractures as contextualized to the specific organizational context, and provides the input to the process specification, which involves clinicians and managers who formally build and annotated the relevant BPMN model.

In the example application case, the data needed to annotate the BPMN model have been collected from the information system (i.e., database and log files) of an Italian hospital, assuming that clinicians and managers are usually provided with such information. Specifically, the tasks duration (i.e., the time required to complete each task in the process) and the patient arrival rate have been retrieved for a one year time interval. According to literature [25], task duration and patient arrival rate have been modeled as values of Exponential and Poisson distributions, respectively.

A preliminary validation of the simulation model has been carried out by verifying that the obtained end-to-end processing time corresponds to the *length of stay* data provided by the hospital information system.

In the simulated scenario different kinds of process performance indicators have been considered. The *timing to surgery* has been used as a disease-specific indicator and set to be lower than 48 hours, according to acknowledged evidence-based guidelines that state that patients with this diagnosis should be operated within two days from admission [28].

Moreover, in order to assess the causes that could lead to pre-operation delays and hinder the effective management of hip fracture patients, other recognized general performance indicators have been assessed, such as the turn-around time of X-ray and Lab tests and the ED patient throughput time.

From the collected data, we noticed that the turn-around time of X-ray and Lab tests requested by ED almost satisfy the general guidelines, being around 60 minutes. Table I shows the target and score (i.e., actual) values for timing to surgery and ED throughput time. The *score* column shows a significant delay in time to surgery (only 64% of patients have been operated within two days from admission), while the ED patient throughput time has been found to satisfy the guidelines (i.e., 98% of patients have been discharged in less than 3 hours).

In this respect, the simulation-based analysis has been focused on the pre-operation phase, in order to find room for improvement.

B. Simulation-based analysis

The annotated BPMN model of Figure 2 has been used to generate the corresponding eBPMN code, according to the method illustrated in Section III.

TABLE I. ACTUAL SCORE AGAINST KPIs

KPI	Target	Score
Timing to Surgery	≤ 48 hours from admission	64%
ED throughput time	≤ 3 hours from admission	98%

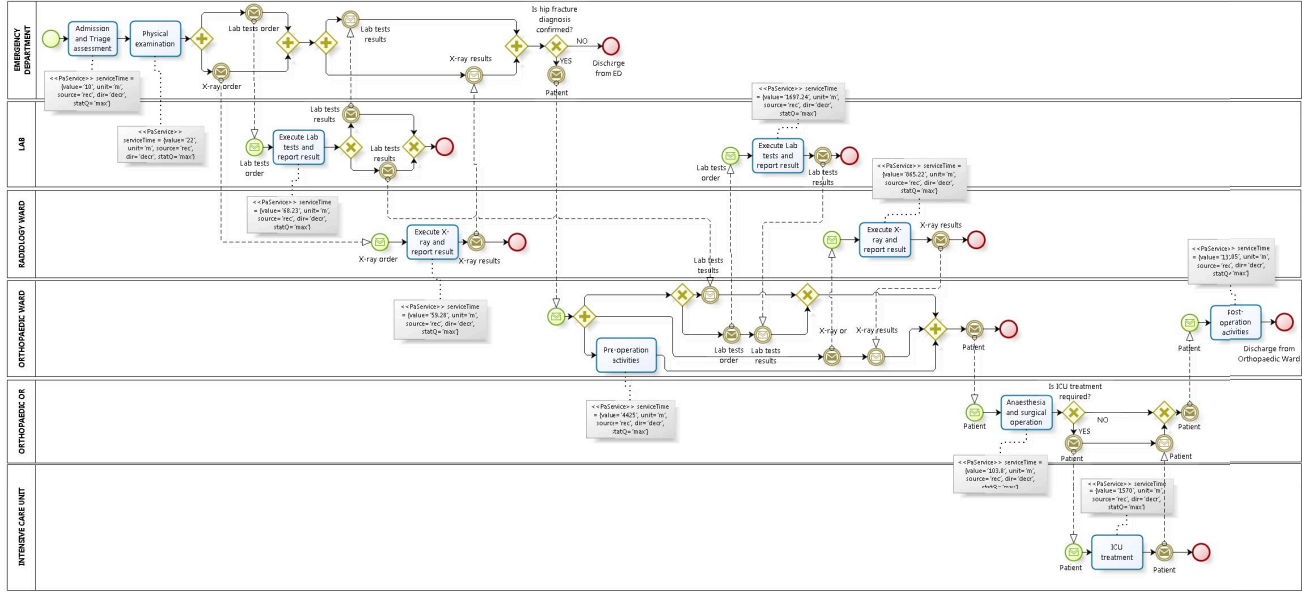


Fig. 2. Annotated BPMN for the considered hip fracture HP.

The following listing gives a portion of the obtained code, namely the eBPMN code corresponding to the *Admission and Triage Assessment* task with the parameters of the associated resource.

```
SimpleResource resource_Admission_and_Triage = new
SimpleResource(new EBPMNName("
Resource_Admission_and_Triage"), new
ExponentialCumulativeFunction(0.0), new
ExponentialStream(new JavaSimGenerators(),
1.0/10),1, timeFactory, layer2Factory);

Task_Admission_and_Triage = new TaskWithResources(
new EBPMNName("Task_Admission_and_Triage"),
EBPMNTimeFactory.makeFrom(Time.ZERO),
timeFactory, layer2Factory);

Task_Admission_and_Triage.addResource(
resource_Admission_and_Triage);
```

A *SimpleResource* object is created and relevant attributes are specified (i.e., *ExponentialStream* implements the exponential distribution modelling the *Admission and Triage service time* of 10 minutes). Next, a *TaskWithResources* object for the *Admission and Triage* task is created and associated to the aforementioned resource.

The complete eBPMN code is then executed on top of the *SimArch* simulation engine in order to get the simulation results that provide the prediction of the relevant performance properties for the considered *HP*.

The obtained results show that the turnaround-time of Lab tests, in case of requests made during the pre-operation activities, could represent a bottleneck, thus causing significant delays that impact the time to surgery.

In order to demonstrate the effectiveness of the proposed method, a *what-if analysis* has been carried out by simply

varying the patients arrival rate and by taking into account both the current scenario and an alternative scenario in which the resources for conducting Lab tests are doubled.

Repeated simulation runs lead to the results depicted in Figure 3, which gives the mean time to surgery (in hours) with respect to the patient inter-arrival mean time (in hours) for both the current scenario (denoted as *Conf 1*) and the aforementioned alternative scenario (denoted as *Conf 2*).

Figure 3 shows that, with the actual patient inter-arrival mean time (61 hours), the simulation of the current scenario yields a time to surgery value of 57 hours, thus greater than the constraint of max 48 hours (see Table I). With this scenario, the 48 hours constraint would be satisfied only in case of a patient inter-arrival time of 107 hours, which is not compliant to the actual inter-arrival time (i.e., the one measured on the real system).

The alternative scenario shows instead that the max 48 hours time to surgery constraint is met by a patient inter-arrival time not lower than 29 hours, a value that is largely satisfactory with respect to the actual patient inter-arrival time.

From a cost-benefit perspective, the proposed approach reduces the modeling and simulation effort while increasing the analysis effectiveness through the full involvement of process stakeholders. Moreover, the use of simulation-based quantitative techniques gives significant advantages with respect to the use qualitative analysis techniques in terms of a more accurate evaluation of the trade-off between the potential performance improvement and the costs associated to the required resources.

A similar analysis can be easily carried out for evaluating the *HP* ability to satisfy other relevant KPIs, without being required to have specific skills in terms of simulation languages

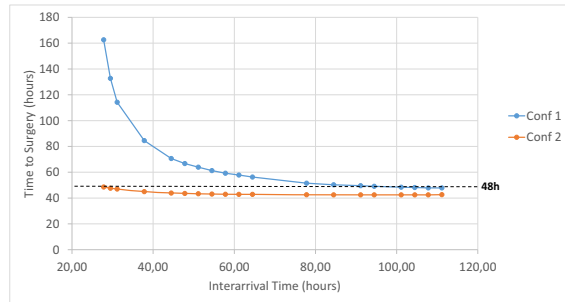


Fig. 3. Time to Surgery vs. Interarrival Time for alternative configurations,

and tools, but simply annotating the BPMN model with relevant performance properties and constraints.

V. CONCLUSIONS

This paper has introduced a model-driven method to enact the simulation-based analysis of *HPs*. The proposed method is founded on two automated model transformations that take as input the BPMN model of the process and yields as output the corresponding simulation code, ready to be executed. The BPMN model is appropriately enriched with performance parameters by use of the PyBPMN extension, while the simulation code is produced in eBPMN, a domain-specific language for business process simulation.

The proposed approach is largely automated and can be effectively used by process stakeholders (e.g., clinicians or healthcare managers) who do not have specific expertise in the fields of simulation-based performance analysis. In addition, the extensive use of model-driven techniques also provides a significant degree of customizability, which facilitates the tailoring of the proposed model transformations to specific needs (e.g., to make use of alternative simulation platforms or to enforce the expressiveness of PyBPMN).

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